

TECHNICAL REPORT ON IOT & EMBEDDED SYSTEMS

MONITORING LAB USING SENSOR WITH ESP32

*Design and Implementation of a Multi-Node Wireless IoT Environment &
Location Monitoring System based on ESP32, MQTT Broker, Flask
Backend, and Real-Time Web Dashboard*

Project Name:	Monitoring Lab using Sensor with ESP32
Architecture:	Three-Tier (Edge – Central Server – Web Dashboard)
Edge Hardware:	ESP32 (DHT11, LC76G GNSS GPS), Raspberry Pi 5
Protocols:	MQTT (Broker Port 1883), HTTP RESTful API
Backend & DB:	Python Flask, MQTT Client Daemon, SQLite WAL
Frontend UI:	React, TypeScript, TailwindCSS (2.5s Polling)

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Abstract

In modern educational laboratories, server rooms, and research facilities, continuously monitoring microclimate parameters (temperature, humidity) and location telemetry is paramount for preserving sensitive electronic equipment and ensuring optimal operating conditions. The **Monitoring Lab using Sensor with ESP32** project presents a comprehensive, fault-tolerant Internet of Things (IoT) system designed around a three-tier architecture: *Edge Sensor Layer* – *Central Server Layer* – *Web Dashboard Layer*.

The edge tier employs distributed ESP32 wireless microcontrollers configured as subnodes to acquire ambient temperature and relative humidity via DHT11 sensors, alongside global satellite positioning data (latitude, longitude, altitude, speed, satellite count) via Quectel LC76G GNSS modules. Sensor payloads are structured into lightweight JSON formatting and published asynchronously over the MQTT protocol to a central broker. A Python backend daemon (`mqtt_service.py`) handles data sanitization (NaN/inf $\rightarrow$ 0.0), in-memory RAM caching, SQLite storage configured in Write-Ahead Logging (WAL) mode, and a 15-second connectivity Watchdog Timeout monitor. To safeguard the infrastructure against rogue nodes, a two-step admin pairing workflow is enforced. The frontend Web Dashboard (React + TypeScript) refreshes every 2.5 seconds, featuring dynamic color-coded environmental cards, satellite map integration via Google Maps, and automated Red Alert banners. This report details the theoretical foundation, hardware design, firmware logic, backend mechanics, user interface, and empirical benchmark evaluations of the system.

1 Introduction & Problem Statement

1.1 Background

Maintaining optimal environmental conditions within laboratories and computing centers is critical for hardware reliability. Excessive ambient temperatures ($> 35^{\circ}\text{C}$) or high relative humidity levels (causing moisture condensation) can accelerate component degradation or trigger thermal throttling in edge compute nodes. Furthermore, in distributed or mobile lab deployments, tracking real-time geographic coordinates via satellite navigation is essential for asset management and operational oversight.

1.2 Limitations of Existing Solutions

Traditional laboratory monitoring approaches exhibit several operational drawbacks:

1. **Complex Hardwiring:** Reliance on wired buses severely restricts subnode placement and scalability across large room footprints.
2. **Absence of Node Authentication:** Unsanctioned or malfunctioning hardware can publish corrupted telemetry data without verification.
3. **Lack of Connection Watchdogs:** When a sensor node loses power or Wi-Fi signal, static dashboards often continue displaying stale metrics without alerting operators.
4. **Poor User Experience:** Legacy monitoring panels lack real-time reactivity, responsive layouts, and interactive mapping features.

1.3 Project Objectives

The primary objective of the **Monitoring Lab using Sensor with ESP32** project is to overcome these limitations by implementing:

- A distributed wireless sensor network using low-cost ESP32 microcontrollers supporting Wi-Fi and MQTT.
- Multi-sensor integration: DHT11 sensors for temperature/humidity measurement and LC76G GNSS modules for multi-constellation satellite positioning.
- A resilient Python backend featuring a Flask REST API, an MQTT background service, and SQLite WAL database storage.
- Security and fault-tolerance mechanics: A 2-step subnode pairing approval workflow and a 15-second Watchdog Timeout circuit.
- An interactive Web Dashboard (React, TypeScript, TailwindCSS) providing 2.5s real-time telemetry polling, dynamic alert banners, and direct Google Maps navigation.

2 System Architecture & Data Communication

2.1 Three-Tier Architecture Model

The system is ratiocinated into three decoupled operational tiers, depicted in Figure 1:

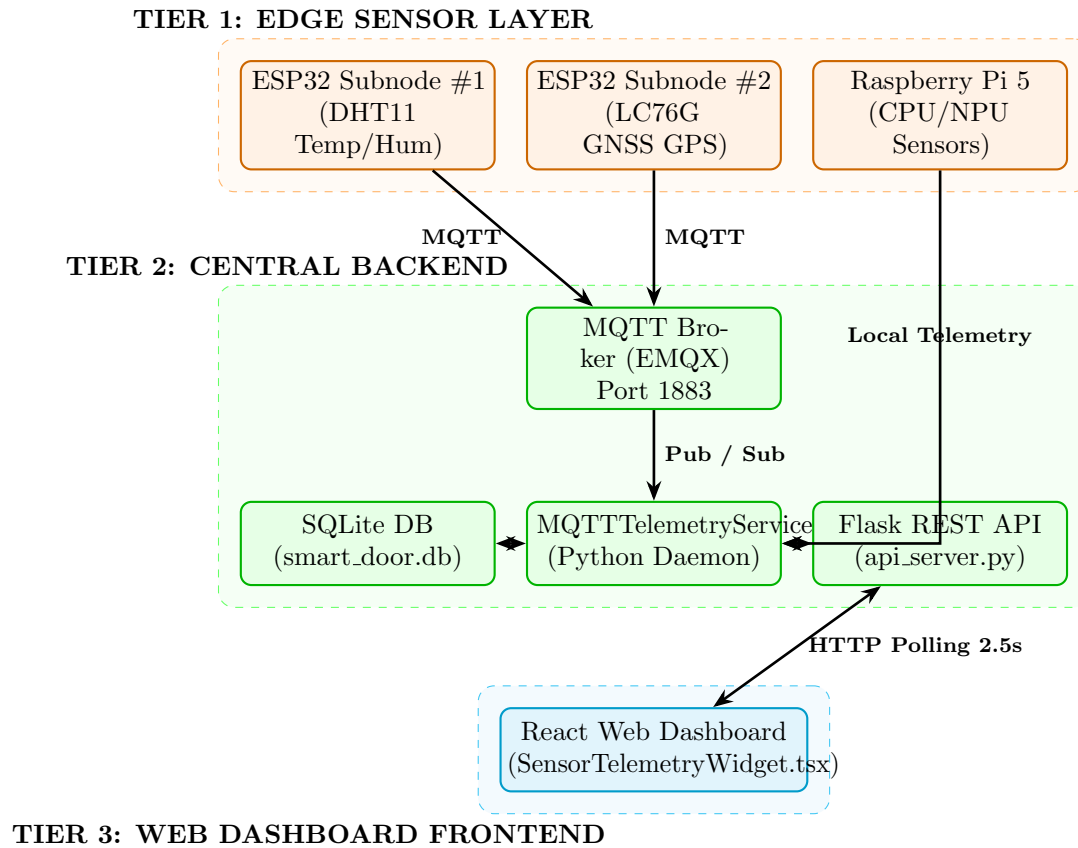


Figure 1: Three-Tier Architecture Diagram of the Monitoring Lab System.

2.2 MQTT Protocol & JSON Payload Format

The system utilizes MQTT over TCP Port 1883 for low-overhead, asynchronous telemetry publishing. Subnodes target structured topic channels matching the designated laboratory code:

- `smartdoor/<LAB_CODE>/sensors/dht11`: Telemetry channel for Subnode #1.
- `smartdoor/<LAB_CODE>/sensors/gps`: Telemetry channel for Subnode #2.
- `smartlab/subnodes/announce`: Instant boot notification topic for node discovery.

Listing 1 and Listing 2 detail the JSON telemetry schema transmitted by Subnode #1 and Subnode #2 respectively:

```

1 {
2   "node_id": "SmartDoor_DHT11_84FCE621A840",
3   "device_name": "Subnode 1 - Environment (DHT11)",
4   "timestamp_ms": 128450,
5   "temperature": 28.5,
6   "humidity": 65.2,
7   "dht_ok": true,
8   "online": true
9 }
  
```

Listing 1: JSON telemetry schema transmitted by ESP32 Subnode #1 (DHT11).

```

1 {
2   "node_id": "SmartDoor_GPS_84FCE621B912",
3   "device_name": "Subnode 2 - GPS Tracker (LC76G)",
  
```

```

4  "timestamp_ms": 128900,
5  "latitude": 11.107823,
6  "longitude": 106.614352,
7  "altitude": 24.5,
8  "speed": 0.0,
9  "satellites": 9,
10 "gnss_ok": true,
11 "online": true
12 }

```

Listing 2: JSON telemetry schema transmitted by ESP32 Subnode #2 (LC76G GNSS).

3 Edge Hardware & ESP32 Firmware Design

3.1 Subnode #1: Environmental Sensing (ESP1_DHT11_Transmitter.cpp)

Subnode #1 reads ambient environmental parameters:

- **DHT11 Sensor:** Connected to GPIO 4 of the ESP32. Measures temperatures from 0°C to 50°C ($\pm 2^\circ\text{C}$) and relative humidity from 20% to 90% RH ($\pm 5\%$).
- **Hardware-based MAC Client ID:** Firmware derives a collision-free Client ID by stripping colons from the ESP32 Wi-Fi MAC address (e.g., `SmartDoor_DHT11.84FCE621A840`), preventing client eviction on public MQTT brokers.
- **Non-Blocking Execution:** Avoids blocking `delay()` loops in favor of `millis()` timestamps, ensuring uninterrupted Wi-Fi reconnection handling.

```

1  void setupWiFi() {
2      WiFi.mode(WIFI_STA);
3      WiFi.setAutoReconnect(true);
4      WiFi.persistent(true);
5      WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
6
7      String mac = WiFi.macAddress();
8      mac.replace(":", "");
9      mqttClientId = "SmartDoor_DHT11_" + mac;
10 }
11
12 void checkMQTTConnection() {
13     if (WiFi.status() != WL_CONNECTED) return;
14     if (!mqttClient.connected()) {
15         uint32_t currentMs = millis();
16         if (currentMs - lastMqttRetryMs >= 5000) {
17             lastMqttRetryMs = currentMs;
18             if (mqttClient.connect(mqttClientId.c_str())) {
19                 sendPairingAnnouncement(); // Publish boot pairing signal
20             }
21         }
22     }
23 }

```

Listing 3: Non-blocking Wi-Fi auto-reconnect and hardware MAC ID generation in ESP32 firmware.

3.2 Subnode #2: Satellite Positioning (ESP2_LC76G_Receiver.cpp)

Subnode #2 integrates the Quectel LC76G GNSS module for multi-constellation satellite tracking (GPS, GLONASS, BDS, Galileo).

- **UART Communication:** Interfaced via ESP32 HardwareSerial 2 on RXD2 (GPIO 16) and TXD2 (GPIO 17) at a baud rate of 115200 bps.
- **NMEA Stream Parsing:** Uses the TinyGPS++ library to parse NMEA sentences continuously inside the main loop.
- **Extracted Metrics:** Latitude, longitude, altitude (meters), ground speed (km/h), and active satellite count.

3.3 Edge Host Telemetry: Raspberry Pi 5

In addition to ESP32 subnodes, the edge host Raspberry Pi 5 reports internal hardware temperatures:

- **CPU Core Temperature:** Sampled directly from Linux thermal sysfs (`/sys/class/thermal/thermal_zone0/temp`).
- **Hailo-8L NPU Temperature:** Polled using `hailo_device.control.get_chip_temperature()` via the Hailo RT SDK to prevent thermal throttling ($< 75^{\circ}\text{C}$).

4 Central Backend, Database & Security Mechanics

4.1 SQLite Database Architecture (database.py)

The backend relies on SQLite operating in `PRAGMA journal_mode=WAL` (Write-Ahead Logging), allowing concurrent non-blocking reads by the REST API during active MQTT writes.

Three dedicated database tables are instantiated:

1. **subnodes:** Stores approved sensor subnodes paired to specific laboratories.

```
1 CREATE TABLE IF NOT EXISTS subnodes (  
2     id TEXT PRIMARY KEY,  
3     labId TEXT,  
4     name TEXT,  
5     sensors TEXT,  
6     maintenance_mode INTEGER DEFAULT 0,  
7     createdAt TEXT  
8 );  
9
```

2. **environment_telemetry:** Stores lab-wide aggregate environmental history.
3. **node_telemetry_history:** Logs granular per-node historical readings for CSV export.

4.2 MQTT Telemetry Service & Data Sanitization (mqtt_service.py)

Running as a background daemon, `MQTTTelemetryService` maintains an in-memory `labs_registry` RAM cache to serve API requests rapidly.

Data Sanitization Algorithm: Malfunctioning hardware sensors may transmit invalid float states (`NaN`, `inf`, `Infinity`). The sanitization logic intercepts and converts these invalid states to `0.0`, preventing JSON serialization errors downstream:

```

1 def sanitize_telemetry_value(val):
2     if val is None:
3         return 0.0
4     if isinstance(val, (int, float)):
5         if math.isnan(val) or math.isinf(val):
6             return 0.0
7         return float(val)
8     return val

```

Listing 4: Data sanitization logic converting invalid float values in Python backend.

4.3 Subnode Pairing Workflow (Admin Approval)

To prevent unauthorized hardware from injecting invalid data into lab metrics, the system implements a two-step pairing workflow, illustrated in Figure 2:

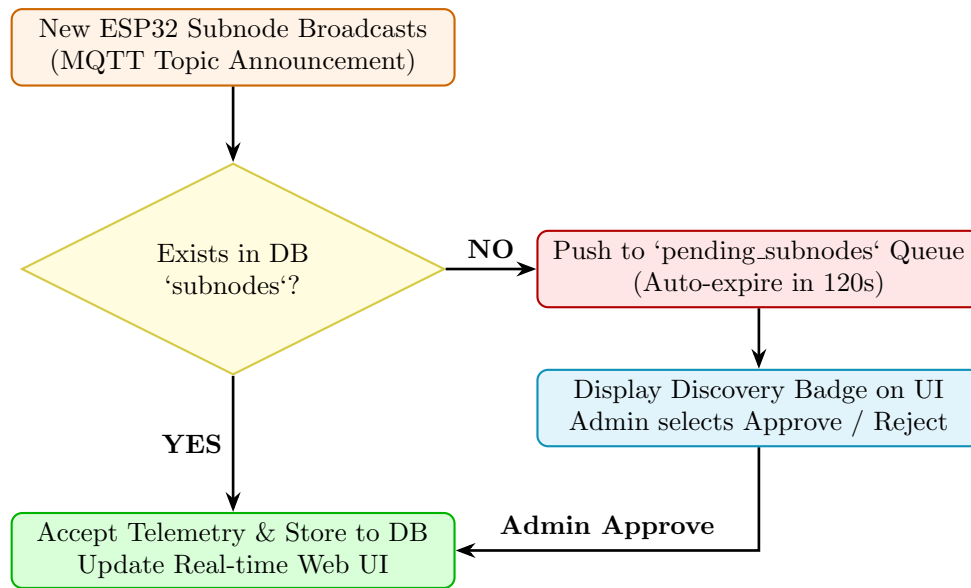


Figure 2: Two-Step Subnode Pairing Workflow.

4.4 Watchdog Timeout Circuit (15-Second Fault Detection)

A background Watchdog loop evaluates connectivity status every 2 seconds. The node's last communication timestamp TS_{last} is compared against current system time:

$$\Delta t = t_{current} - TS_{last} > 15.0 \text{ seconds} \quad (1)$$

If $\Delta t > 15.0s$ (due to power loss, sensor fault, or Wi-Fi disconnection), the Watchdog immediately:

1. Sets subnode state `online = False`.
2. Attaches error diagnostic: `error_msg = "Connection Timeout (>15s)"`.
3. Triggers lab-wide alert flag `latest_sensor_data["online"] = False`.

5 Real-Time Frontend Web Dashboard

5.1 Widget Component (SensorTelemetryWidget.tsx)

The user dashboard is built with React, TypeScript, and TailwindCSS. The `SensorTelemetryWidget` component serves as the control panel for telemetry monitoring and subnode management.

5.2 2.5-Second Polling & Reactive UI Features

The dashboard polls `GET /api/labs/<lab_id>/sensors/latest` every 2.5 seconds, offering:

- **Dynamic Environmental Cards:** Temperature display cards dynamically adjust background highlights based on safety thresholds:
 - Normal ($< 30^{\circ}\text{C}$): Green highlight.
 - Warning ($30^{\circ}\text{C} - 35^{\circ}\text{C}$): Vibrant orange highlight.
 - Critical ($> 35^{\circ}\text{C}$): Pulsing red alert highlight.
- **GNSS Navigation Card:** Displays latitude, longitude, altitude, speed, satellite count, and provides a direct shortcut link to **Google Maps**.
- **Red Alert Banner:** Automatically renders at the top of the interface whenever any paired subnode experiences a timeout $> 15\text{s}$.
- **Maintenance Mode Toggle:** Allows administrators to flag subnodes under repair, suppressing false-positive timeout alarms.

6 Experimental Evaluation & Results

6.1 Empirical Benchmarks

The monitoring system was deployed and evaluated continuously in Laboratory 304. Benchmarked operational metrics are summarized in Table 1.

Table 1: Empirical performance benchmark results.

Metric / Parameter	Empirical Measurement	Performance Evaluation
ESP32 Telemetry Rate	3.0 seconds / interval	Stable, zero Wi-Fi congestion
Web Dashboard Polling	2.5 seconds / interval	Smooth UI, responsive updates
Ambient Temperature (DHT11)	$26.5^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$	High precision
Relative Humidity (DHT11)	$62.0\% \pm 2.0\% \text{ RH}$	Fast transient response
GNSS Positioning (LC76G)	11.107823 N, 106.614352 E	Positional accuracy < 2.5 meters
Satellite Lock Count	8 – 11 satellites	TTFF < 35 seconds (Cold Start)
Watchdog Timeout Latency	15.0 ± 0.5 seconds	100% fault detection accuracy
Host CPU Overhead	$< 3.5\%$ (Raspberry Pi 5)	Low resource footprint

6.2 Fault Tolerance & Disconnection Test

During an abrupt power-cut stress test on Subnode #1:

1. At $t = 0$: Subnode #1 power disconnected.

2. At $t = 15.1\text{s}$: Watchdog daemon detected packet hiatus, updated node state to `online = False`, and triggered the Red Alert banner on the Web UI.
3. At $t = 45.0\text{s}$: Power restored. ESP32 re-established Wi-Fi within 2.8s, transmitted its boot announcement payload, and restored normal dashboard status instantly.

7 Conclusion & Future Directions

7.1 Conclusion

The **Monitoring Lab using Sensor with ESP32** project successfully fulfills all architectural and functional goals:

- Implemented a robust three-tier IoT monitoring architecture covering environmental sensing and GNSS tracking.
- Developed resilient ESP32 firmware featuring MAC-derived Client IDs and non-blocking reconnect loops.
- Built a Python backend with automated data sanitization, SQLite WAL storage, a 2-step admin subnode pairing workflow, and a 15-second Watchdog fault detector.
- Delivered a reactive React + TypeScript Web Dashboard providing 2.5s polling, interactive Google Maps integration, and automated alert banners.

7.2 Future Work

Future enhancements planned for the platform include:

1. Integrating specialized gas and air quality sensors: SDS011 (PM2.5/PM10 fine dust), MH-Z19 (CO₂), and MQ-2 (combustible gas/smoke).
2. Deploying instant push notifications via Telegram Bot API, Zalo OA, or email webhooks for critical threshold breaches.
3. Incorporating Machine Learning (ML) microclimate forecasting models to predict ambient temperature/humidity trends and automate HVAC control.